

INTERNATIONAL STANDARD



**Radio-frequency connectors –
Part 52: Sectional specification for series MMCX RF coaxial connectors**



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RADIO-FREQUENCY CONNECTORS –

Part 52: Sectional specification for series MMCX RF coaxial connectors

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The text of this standard is based on the following documents:

FDIS	Report on voting
46F/314/FDIS	46F/320/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts of the IEC 61169 series, under the general title: *Radio-frequency connectors*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

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- withdrawn,
- replaced by a revised edition, or
- amended.

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RADIO-FREQUENCY CONNECTORS –

Part 52: Sectional specification for series MMCX RF coaxial connectors

1 Scope

This part of IEC 61169, which is a sectional specification (SS), provides information and rules for the preparation of detail specifications (DS) for RF coaxial connectors with snap-on coupling, typically for use in 50 Ω cable networks (MMCX).

It prescribes mating face dimensions for general purpose connectors – grade 2, dimensional details of standard test connectors-grade 0, gauging information and tests selected from IEC 61169-1, applicable to all detail specifications relating to series MMCX RF connectors.

This specification indicates recommended performance characteristics to be considered when writing a detail specification and it covers test schedules and inspection requirements for assessment levels M and H.

The MMCX miniature snap-on coupling structure series R.F. coaxial connector with the characteristic of normative impedance 50 Ω are used with various kinds of R.F cables and strips. The operating frequency limit is up to 6 GHz.

NOTE Metric dimension are original dimensions. All undimensioned pictorial configurations are for reference purpose only.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61169-1:2013, *Radio frequency connectors – Part 1: Generic specification – General requirements and measuring methods*

IEC 60068-1, *Environmental testing – Part 1: General and guidance*

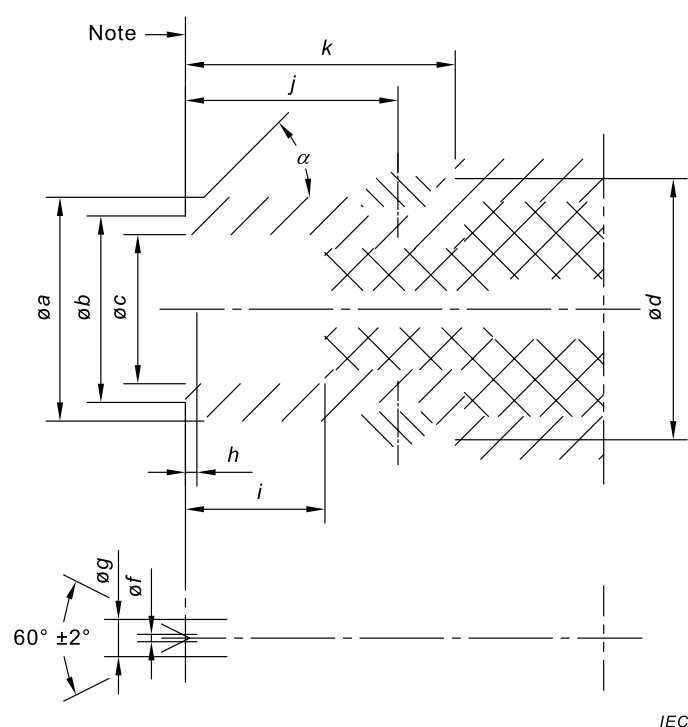
IEC 62037 (all parts), *Passive RF and microwave devices, intermodulation level measurement*

3 Mating face and gauge information

3.1 Dimensions – General connectors – Grade 2

3.1.1 Connector with pin centre contact

The mating face of connector with pin centre contact is shown in Figure 1 and its dimensions are shown in Table 1.



NOTE Mechanical and electrical reference plane.

Figure 1 – Connector with pin centre contact

Table 1 – Dimensions of connector with pin centre contact

Ref.	mm	
	Min.	Max.
<i>a</i>	2,35	2,40
<i>b</i>	1,80	--
<i>c</i>	1,58	1,62
<i>d</i>	--	2,83
<i>f</i>	--	0,20
<i>g</i>	0,38	0,42
<i>h</i>	0,00	0,25
<i>i</i> ^a	1,23	--
<i>j</i> ^b	2,08	2,12
<i>k</i>	2,70	--
α	30°	--
^a The dimension should meet the mechanical and electrical requirements, also compensate the electrical effect caused by gaps. ^b The shape of elastic gasket is optional and its diameter and position should meet the mechanical and electrical requirements.		

3.1.2 Connector with socket centre contact

The mating face of connector with socket centre contact is shown in Figure 2 and its dimensions are shown in Table 2.

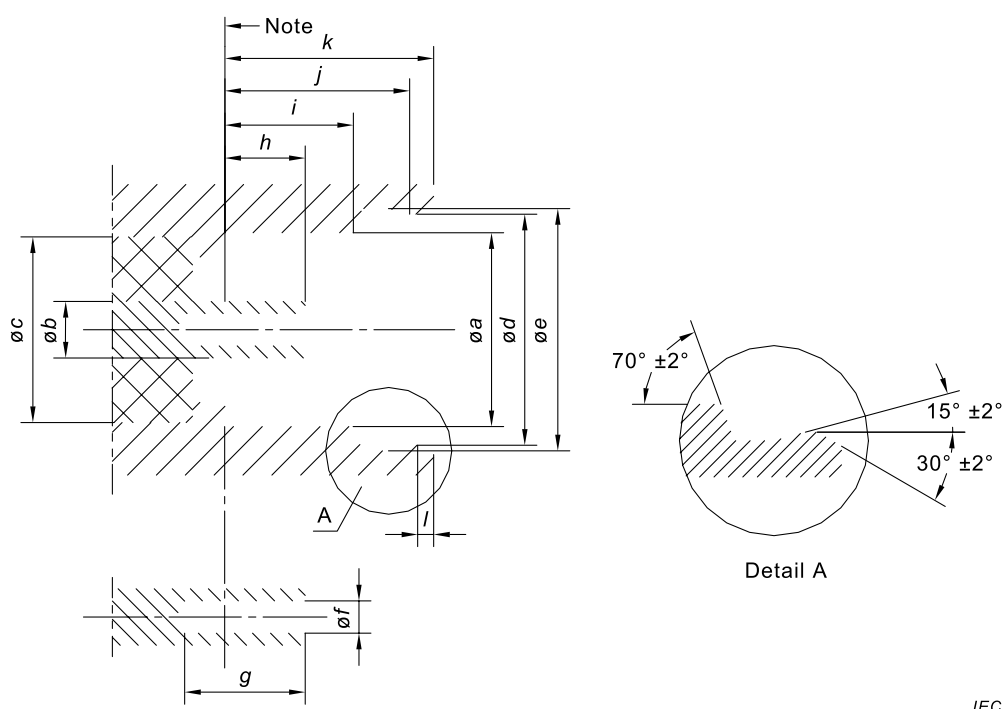


Figure 2 – Connector with socket centre contact

Table 2 – Dimensions of connector with socket centre contact

Ref.	mm	
	Min.	Max.
<i>a</i>	2,41	--
<i>b</i>	0,70 (nominal)	
<i>c^a</i>	--	--
<i>d</i>	2,88	2,92
<i>e</i>	3,00	3,04
<i>f^b</i>	--	--
<i>g</i>	1,40	--
<i>h</i>	0,90	1,20
<i>i</i>	1,57	1,63
<i>j</i>	2,26	2,34
<i>k</i>	--	2,69
<i>l</i>	0,14	0,23
^a The diameter chosen shall meet nominal impedance of 50 Ω and electrical and mechanical requirements.		
^b Centre contact design is optional. It should however meet the gauging requirements of 3.2.1 and relevant reflection factor requirements of Clause 4 using grade 0 connector in accordance with Figure 5.		

3.2 Gauges

3.2.1 Gauge pin for socket centre contact

The gauge pin for socket centre contact is shown in Figure 3 and its dimensions are shown in Table 3.

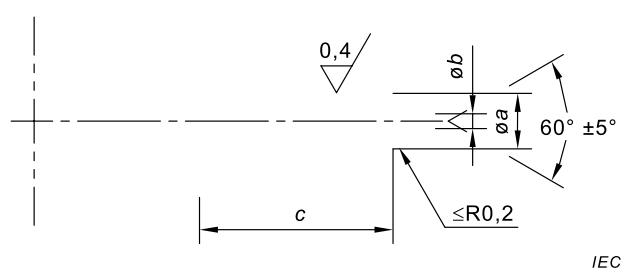


Figure 3 – Gauge pin for socket centre contact

Table 3 – Dimensions of gauge pin for socket centre contact

Ref.	Gauge A (For sizing purpose)		Gauge B (For retention purpose) Mass of gauge : 28^{+2}_0 g	
	mm		mm	
	Min.	Max.	Min.	Max.
<i>a</i>	0,420	0,425	0,375	0,380
<i>b</i>	0,00	0,20	0,00	0,20
<i>c</i>	0,90	1,10	1,00	1,20
Material: steel, polished.				

Test procedure:

a) Sizing test

The gauge A should insert the socket centre contact three times, this is a sizing operation.

b) Insertion test

Following the sizing operation, if specified in the detail specification, the force necessary to insert gauge A fully into the socket centre contact shall be measured. When this test is required, the insertion force shall not exceed 5 N.

c) Retention test

After sizing or insertion test, the gauge B shall be inserted into the socket centre contact. The contact shall retain the mass of the gauge C in a vertical downward position.

3.2.2 Gauge for outer contact

The gauge for outer contact is shown in Figure 4 and its dimensions are shown in Table 4.

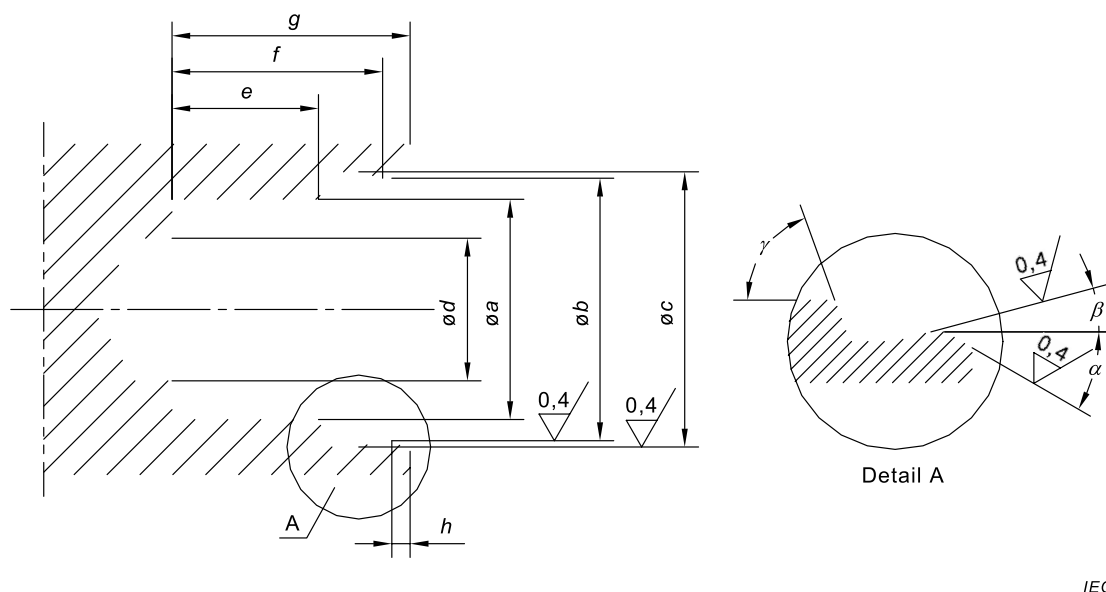


Figure 4 – Gauge for outer contact

Table 4 – Dimensions of gauge for outer contact

Ref.	Gauge A (For sizing purpose)		Gauge B (For retention purpose) Mass of gauge : 600 ⁺⁵ ₀ g	
	mm		mm	
	Min.	Max.	Min.	Max.
a	2,41	2,425	2,42	2,425
b	2,875	2,88	2,92	2,925
c	3,04	3,05	2,99	3,00
d	1,58	1,62	1,58	1,62
e	1,63	1,64	1,56	1,57
f	2,30	2,305	2,255	2,26
g	2,65	2,66	2,65	2,66
h	0,135	0,14	0,13	0,23
α	32°	32,5°	27°	33°
β	15,5°	16°	14°	14,5°
γ	69°	71°	69°	71°
Material: steel, polished.				

Test procedure:

a) Sizing test

The gauge A should insert the outer contact once, this is a sizing operation.

b) Insertion test

Following the sizing operation, if specified in the detail specification, the force necessary to insert gauge A fully into the outer contact shall be measured. When this test is required, the insertion force shall not exceed 18 N.

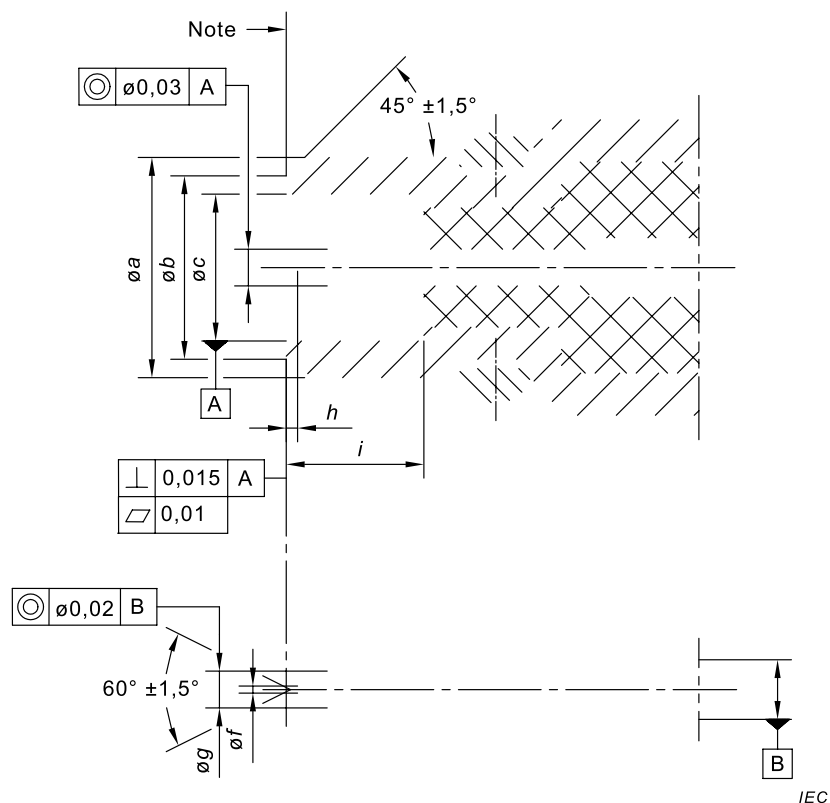
c) Retention test

After sizing or insertion test, the gauge B shall be inserted into the outer contact. The contact shall retain the mass of the gauge B in a vertical downward position.

3.3 Dimensions – Standard test connectors – Grade 0

3.3.1 Connector with pin centre contact

The mating face of standard test connector with pin centre contact is shown in Figure 5 and its dimensions are shown in Table 5.



NOTE Mechanical and electrical reference plane.

Figure 5 – Connector with pin centre contact

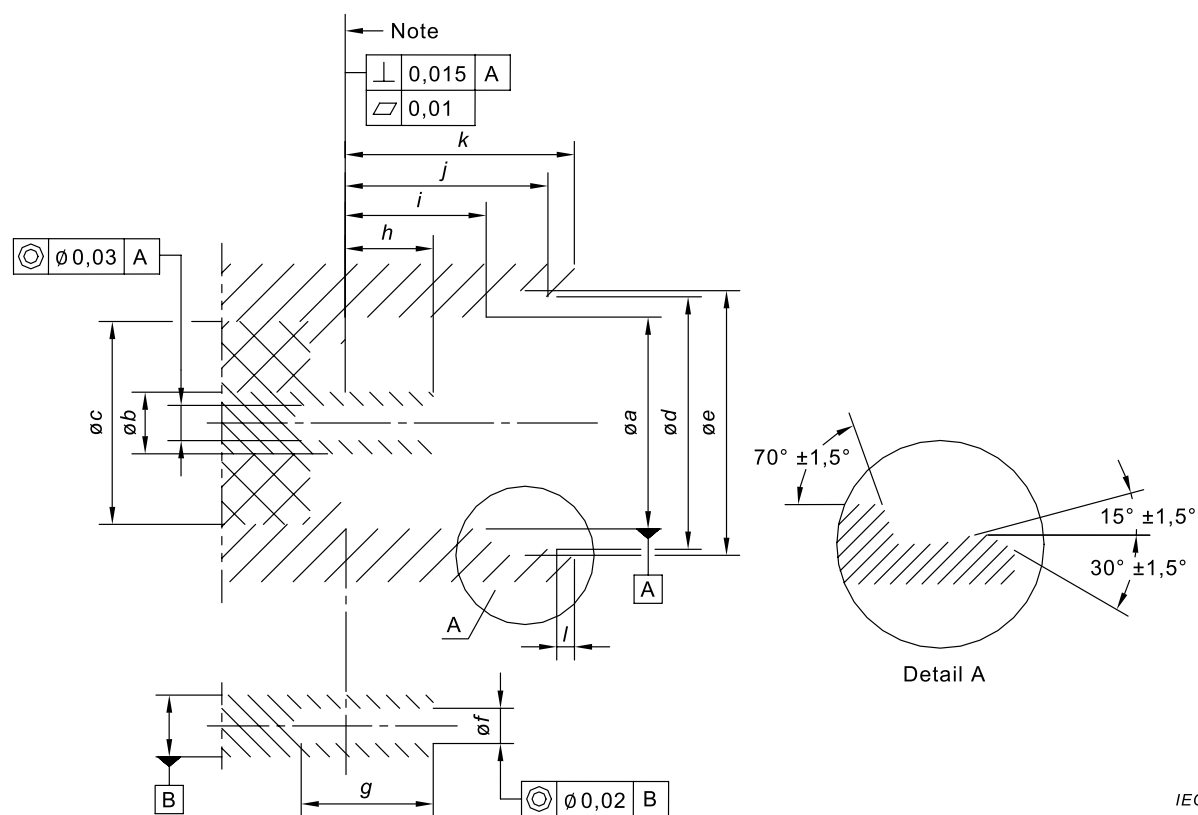
Table 5 – Dimensions of connector with pin centre contact

Ref.	mm	
	Min.	Max.
a	2,38	2,405
b	2,10	2,20
c	1,59	1,61
f	0,10	0,20
g	0,39	0,41
h	0,10	0,25
i	1,23	1,55

NOTE Dimensions not indicated in Table 5 are included in Table 1.

3.3.2 Connector with socket centre contact

The mating face of standard test connector with socket centre contact is shown in Figure 6 and its dimensions are shown in Table 6.



NOTE Mechanical and electrical reference plane.

Figure 6 – Connector with socket centre contact

Table 6 – Dimensions of connector with socket centre contact

Ref.	mm	
	Min.	Max.
<i>a</i>	2,41	2,42
<i>b</i>	0,69	0,71
<i>c^a</i>	--	--
<i>d</i>	2,88	2,92
<i>e</i>	3,01	3,03
<i>f^b</i>	--	--
<i>g</i>	1,40	--
<i>h</i>	1,00	1,20
<i>i</i>	1,57	1,63
<i>j</i>	2,26	2,34
<i>k</i>	--	2,69
<i>l</i>	0,14	0,23

^a The diameter chosen shall meet the characteristic impedance of $(50 \pm 0,5) \Omega$.

^b Design for slotting is optional, and should meet electrical and mechanical requirements, when mating with $\Phi 0,39 \text{ mm} \sim \Phi 0,41 \text{ mm}$ gauge pin.

4 Quality assessment procedure

4.1 General

Clause 10 of IEC 61169-1:2013 applies to this sectional specification, except the following requirement:

Subclauses 4.2 to 4.4 provide recommended rating, performance and test conditions to be considered when writing a detail specification. They also provide an appropriate schedule of tests with minimum levels of conformance inspection sampling. Together with the pro forma blank detail specification (BDS) and instructions for the preparation of a detail specification.

4.2 Rating and characteristics (see Clause 5 of IEC 61169-1:2013)

The values indicated below are recommended for MMCX series RF coaxial connectors and are given for the writer of the detail specification. They are applicable for the condition when the connectors are fully mated.

Certain tests are listed without any recommended values being given. These tests will usually not be required. When these tests are required, appropriate values shall be entered in the detail specification at the discretion of the specification writer.

Preferred climatic categories are given in Table 7. Rating and characteristics are given in Table 8.

Table 7 – Preferred climatic categories (see IEC 60068-1)

Category designation	Letter	Temperature range	Damp heat steady state
40/85/21	A	–40 °C to +85 °C	21 days
55/125/21	B	–55 °C to +125 °C	21 days

Table 8 – Rating and characteristics

Rating and characteristics	IEC 61169-1: 2013 Subclause	Value	Remarks deviations from standard test methods
Electrical			
Nominal impedance		50 Ω	
Frequency range ^a		DC ~ 6 GHz	Or upper frequency limit of cable
Return loss ^a	9.2.1		
Grade 2 connectors			
– straight styles		$\geq 20,1$ dB	
– right-angle styles		$\geq 16,8$ dB	
– component mounting styles		See DS	
– solder bucket and PCB mounting styles		See DS	
Power rating	9.2.2	See DS	
Contact resistance, outer conductor and centre conductor continuity (mated cabled connectors)	9.2.3		
Centre contact resistance ^b			
– initial		$\leq 5,0$ m Ω	

Rating and characteristics	IEC 61169-1: 2013 Subclause	Value	Remarks deviations from standard test methods
– after conditioning		$\leq 15,0 \text{ m}\Omega$	
Outer conductor resistance ^b			
– initial		$\leq 2,5 \text{ m}\Omega$	
– after conditioning		$\leq 7,5 \text{ m}\Omega$	
Insulation resistance	9.2.5		
– initial		$\geq 1\,000 \text{ M}\Omega$	
– after conditioning		$\geq 2\,00 \text{ M}\Omega$	
Proof voltage at sea level ^{c d}	9.2.6		(86~106) kPa
– 96IEC50-1 cable		500 V	
– semi-rigid and semi-flexible cable with 0,085 in (2,16 mm) diameter		500 V	
Proof voltage at 4,4 kPa ^{c d}	9.2.6		4,4 kPa approximately equivalent to 20 km
– 96IEC50-1 cable		100 V	
– semi-rigid and semi-flexible cable with 0,085 in (2,16 mm) diameter		100 V	
Environmental test voltage at sea level ^{c d}	9.2.6		(86k~106) kPa
– 96IEC50-1 cable		170 V	
– semi-rigid and semi-flexible cable with 0,085 in (2,16 mm) diameter		250 V	
Environmental test voltage at 4,4 kPa ^{c d}	9.2.6		4,4 kPa approximately equivalent to 20 km
– 96IEC50-1 cable		45 V	
– semi-rigid and semi-flexible cable with 0,085 in (2,16 mm) diameter		45 V	
Screening effectiveness ^e	9.2.7	$\geq 60 \text{ dB at } 1 \text{ GHz}$	$Z_t \leq 50 \text{ m}\Omega$
Mechanical			
Insertion force (resilient contacts)	9.3.4		See 3.2 of this sectional specification
– centre contact		$\leq 5 \text{ N}$	
– outer contact		$\leq 18 \text{ N}$	
Gauge retention force (resilient contacts)	9.3.4		See 3.2 of this sectional specification
– centre contact		$\geq 0,28 \text{ N}$	
– outer contact		$\geq 6 \text{ N}$	
Centre contact captivation	9.3.5		Only applicable to centre contact with captivation structure, after test, the dimensions of centre contact comply with interface dimension
– axial force		10 N	
– permitted displacement in each direction		See DS	
– axial torque		See DS	
Engagement and separation forces and torques	9.3.6		
– engagement		$\leq 18 \text{ N}$	
– separation		$\geq 6 \text{ N}$	

Rating and characteristics	IEC 61169-1: 2013 Subclause	Value	Remarks deviations from standard test methods
Effectiveness of clamping device against cable pulling	9.3.8	See DS	
Effectiveness of clamping device against cable torsion	9.3.10	See DS	
Bump	9.3.13	See DS	
Vibration	9.3.3	100 m/s ² (10~500) Hz	10 g _n acceleration
Shock	9.3.14	See DS	
Environmental			
Salt mist	9.4.10	48 h	Duration of spraying
Endurance			
Mechanical endurance	9.3.15	500 operations	
High temperature endurance ^f	9.4.5	1 000 h at 125 °C	
Low temperature endurance	9.4.6	See DS	
^a These values apply to basic connector. In practice, these may be influenced by the cable used and reference should always be made to the actual values given in the DS. ^b Values for a single pair of connectors. ^c Voltages are r.m.s. values of a.c. at 40 Hz to 65 Hz, unless otherwise specified. ^d Some cables usable with these connectors have ratings lower than the values given here. ^e When interfaces are fully mated. ^f For certain connectors, the upper temperature limit is restricted by the cable characteristics. Reference should be made to the relevant cable specification.			

4.3 Test schedule and inspection requirements

4.3.1 Acceptance tests

Table 9 describes the acceptance tests to be performed.

Table 9 – Acceptance tests

	IEC 61169-1: 2013	Assessment level M (higher)				Assessment level H (lower)			
	Subclause	Test required	IL	AQL	Period	Test required	IL	AQL	Period
				%				%	
Group A1					Lot by lot				Lot by lot
Visual examination	9.1.1	a	II	1		a	S3	1,5	
Group B1									
Outline dimension	9.1.2	a	S4	0,4		a	S3	4,0	
Mechanical compatibility	9.1.2.2	a	II	1		a	S3	1,5	
Engagement and separation	9.3.6	a	S4	0,4		a	S3	1,5	
Gauge retention (resilient contacts)	9.3.4	ia	II	1		ia	S3	1,5	
Sealing									
non hermetic	9.4.7	ia	II	0,65		ia	S3	1	
hermetic	9.4.8	ia	II	0,015		ia	S3	0,025	
Voltage proof	9.2.6	a	II	0,4		a	II	4,0	
Solderability (d)	9.3.2.2	ia	S4	0,4		ia	S3	4,0	
Insulation resistance	9.2.5	a	S4	0,4		a	S3	4,0	
For the symbols, abbreviations and procedures, see the end of Table 10.									

4.3.2 Periodic tests

Table 10 describes the periodic tests to be performed.

There are no group C tests for levels H and M.

Table 10 – Periodic tests

	IEC 61169-1: 2013 Subclause	Assessment level M (higher)				Assessment level H (lower)			
		Test required	Number of specimens	Permitted failures per group1#	Period	Test required	Number of specimens	Permitted failures per group1#	Period
Group D1 (d)			6	1	3 years		3	1	3 years
Solderability connector assemblies	9.3.2.2	ia				ia			
Resistance to soldering heat	9.3.2.3	ia				ia			
Mechanical tests on cable fixing									
cable rotation (nutation)	9.3.7	na				na			
cable pulling	9.3.8	ia				ia			
cable bending	9.3.9	ia				ia			

	IEC 61169-1: 2013 Subclause	Assessment level M (higher)				Assessment level H (lower)			
		Test required	Number of specimens	Permitted failures per group1#	Period	Test required	Number of specimens	Permitted failures per group1#	Period
cable torsion	9.3.10	ia				ia			
Group D2 (d)			6	1	3 years		3	1	3 years
Contact resistance, outer conductor and screen continuity, centre conductor continuity	9.2.3	a				a			
Vibration	9.3.3	a							
Damp heat, steady state	9.4.3	a				a			
Group D3 (d)			1	1	3 years		1*	1	3 years
Dimensions piece-parts and materials	9.1.2	a				a			
Group D4 (d)			6	1	3 years		3	1	3 years
Mechanical endurance	9.3.15	a				a			
High temperature endurance	9.4.5	a				a			
Discharge test	9.2.8								
Climatic conditioning	9.4	na				na			
Group D5 (d)			6	1	3 years		3	1	3 years
Reflection factor	9.2.1	a				a			
Screening effectiveness	9.2.7	a				a			
Water immersion	9.4.9	ia				ia			
Group D6 (d)			6	1	3 years		3	1	3 years
Contact captivation	9.3.5	a				a			
Rapid change of temperature	9.4.4	na				na			
Climatic sequence	9.4.2	a				a			
Group D7 (d)			1§		3 years		1§		3 years
Salt mist	9.4.10	a							

a	Suggested as applicable.
ia	Test suggested (if technically applicable).
na	Not applicable.
IL	Inspection level.
AQL	Acceptable quality level.
*	One set of piece-parts each style and variant, unless using common piece parts.
#	For Qualification Approval (QA), a total of two failures only permitted for level H and 1 failure only for level M from groups D1 to D7.
§	Group D7 – number of pairs for each solvent.
(d)	Destructive tests – specimens shall not be returned to stock.

4.4 Procedures for the qualification approval

4.4.1 Quality conformance inspection

This shall consist of test group A1 and B1 on a lot-by-lot basis and test group D1 to D7 on a periodic basis.

4.4.2 Qualification approval and its maintenance

This shall consist of three consecutive lots passing test groups A1 and B1 followed by selection of specimens from the lots as appropriate. These specimens shall successfully pass the specified periodic D tests.

5 Instructions for preparation of detail specifications (DS)

5.1 General

Detail specifications writers shall use the appropriate blank detail specification (BDS). The following page comprise the BDS dedicated for use with type 50 Ω type MMCX connectors. As such, it will already have entered on it information relating to

- a) the basic specification number applicable to all the detail specifications covering connector styles of the series covered by the sectional specification;
- b) the connector series designation.

The specification writer should enter the details relating to the connector style to be covered as indicated. The number in brackets in the BDS corresponds to the following indications, which shall be given.

5.2 Identification of the component

- 1) Enter the following details:
 - Style: The style designation of the connector include type of fixing and sealing, if applicable.
 - Attachment: By deletion of the inapplicable options of cable/wire given for centre and outer conductors.
 - Special features and marking: As applicable.
- 2) Enter details of assessment level and the climatic category.
- 3) A reproduction of the outline drawing and details of the panel piercing (if applicable). It shall provide the maximum envelope dimensions, also the position of the reference plane and, in the case of a fixed connector, the position of the mounting plane(s) relative to the front face of the connector.

- 4) Any maximum panel thickness limitations for fixed connectors shall be stated.
- 5) Particulars of all variants covered by the DS. As appropriate, the information shall include:
 - cable types (or sizes) applicable to each variant,
 - alternative plated or protective finishes,
 - details of alternative mounting flanges having either tapped or plain mounting holes,
 - details of alternative solder spills or solder buckets including when applicable those for use with microwave integrated circuit (MIC) components.
 - alternative plated or protective finishes;
 - details of alternative mounting flanges having either tapped or plain mounting holes;
 - details of alternative solder spills or solder buckets including, when applicable, those for use with microwave integrated circuit (MIC) components.

5.3 Performances

- 6) Performance data listing the most important characteristics of the connectors in accordance with the requirements of the relevant sectional specification. Deviations from the minimum requirements shall be clearly indicated. Non-applicable parameters shall be marked "na".

5.4 Marking, ordering information and related matters

- 7) Insert marking and ordering information as appropriate, together with details or related documents and any invoked structural similarity.

5.5 Selection of tests, test conditions and severities

- 8) "na" shall be used to indicated non-applicable tests. All tests marked "a" by the detail specification writer shall be mandatory.

When using the normal procedure with a dedicated BDS, the letter "a" – for applicable – shall be entered in the "test required" column against each of the tests indicated as being mandatory in the test schedule of the relevant sectional specification. Any additional test required at the discretion of the specification writer shall also be indicated by an "a".

The specification writer shall also indicate, when necessary, details of deviations from the standard test methods and conditions including any relevant deviations given in the test schedule of the sectional specification.

5.6 Blank detail specification pro-forma for type MMCX connector

The following pages contain the complete BDS pro-forma.

(1)		Page 1 of			
		(2)			
ELECTRONIC COMPONENT OF ASSESSED QUALITY IN ACCORDANCE WITH GENERIC SPECIFICATION IEC 61169-1 NATIONAL REFERENCE		(3) (4) ISSUE			
(5) Detail specification for Radio frequency coaxial connector of assessed quality			Type		
Style		Special features and markings			
Method of cable/wire+ attachment		centre conductor – solder/crimp+ outer conductor – solder/clamp/crimp + + delete as appropriate			
(6) Assessment level. ..	Characteristic impedance ... Ω	Climatic category..../.../.../			
(7) Outline and maximum dimensions		Panel piercing and mounting details			
(8) Variants					
Variant No.	Description of variant	61196 IEC			
.	.	.	.	.	.
.	.	.	.	.	.
Information about manufacturers who have components qualified to this detail specification is available through IECQ on-line certificate system.					

(9) Performance (including limiting conditions of use)

Ratings and characteristics	Variant No. Designation	IEC 61169-1:2013 Subclause	Value	Remarks including any deviations from standard test methods
Electrical				
Nominal impedance			50..Ω	
Frequency range Reflection factor		9.2.1	0 GHz to 7,5 GHz GHz GHz GHz	Measurement frequency range
Centre contact resistance		9.2.3	≤ mΩ ≤ mΩ	Initial After conditioning
Centre conductor continuity		9.2.3	≤ mΩ ≤ mΩ ≤ mΩ ≤ mΩ	Resistance change due to conditioning
Outer contact continuity		9.2.3	≤ mΩ ≤ mΩ	Initial After conditioning
Insulation resistance		9.2.5	≥ GΩ ≥ GΩ	Initial After conditioning
+ Proof voltage at sea level		9.2.6	kV kV kV kV	86 kPa to 106 kPa
+ Proof voltage at 4,4 kPa		9.2.6	kV kV kV kV	kPa (if not 4,4 kPa)
Screening effectiveness		9.2.7	dB at GHz	$Z_t \leq \dots$ mΩ
Discharge test (corona) at sea level		9.2.8	≥ V ≥ V ≥ V ≥ V	Extinction voltage
ADDITIONAL ELECTRICAL CHARACTERISTICS				
Intermodulation level		IEC 62037	dB at GHz	Under 2 carries of +43 dBm
+ Voltage values are r.m.s. values at 50 Hz to 60 Hz, unless otherwise specified.				

Ratings and characteristics		IEC 61169-1:2013 Subclause	Value	Remarks including any deviations from standard test methods
Mechanical				
Soldering		9.3.2		
Gauge retention resilient contacts – inner contact – outer contact		9.3.4	N N	
Centre contact captivation – axial force – permitted displacement in each direction – torque		9.3.5	N mm Nm	
Engagement and separation – axial force and torques		9.3.6		Achievable by hand
Strength of coupling mechanism		9.3.11	N	
Effectiveness of cable fixing against – cable rotation		9.3.7	Rotations	
– cable pulling		9.3.8	N N N N	Point of application and duration mm s mm s mm s mm s
– cable bending		9.3.9	cycles cycles cycles cycles	Length of cable mass mm mm mm mm
– cable torsion		9.3.10	Nm Nm Nm Nm	Duration of applied torque s s s s
Bending moment		9.3.12	Nm	Relative to reference plane
Bumps total		9.3.13	m/s ² to Hz	(g _n acceleration)
Vibration		9.3.3	m/s ² to Hz	(g _n acceleration)
Shock		9.3.14	m/s ² shape ms	(g _n acceleration)
ADDITIONAL MECHANICAL CHARACTERISTICS				

Ratings and characteristics		IEC 61169-1:2013 Subclause	Value	Remarks including any deviations from standard test methods
Environmental				
Climatic category				
Sealing non-hermetically sealed connectors		9.4.7	cm ³ /h	100 kPa to 110 kPa pressure differential
Sealing hermetically sealed connectors		9.4.8	10 ⁻³ bar/cm ³ /h	100 kPa to 110 kPa pressure differential
Water immersion		9.4.9		
Salt mist		9.4.10	h	Duration of spraying
ADDITIONAL ENVIRONMENTAL CHARACTERISTICS				
ENDURANCE				
Mechanical		9.3.15	operations	
High temperature		9.4.5	at h °C	
ADDITIONAL ENDURANCE CHARACTERISTICS				
CHEMICAL CONTAMINATION				
Resistance to solvents and contaminating fluids to be used.		9.4.11		
Applicable fluids				
Sulphur dioxide		9.4.12	days	

(10) Supplementary information

– Marking of the component: in accordance with 11.1 of IEC 61169-1:2013 in the following order of procedure 1) Identity of manufacture 2) Manufacturing date code year /week 3) Component identification variant No./designation Identification
– Marking and contents of package: in accordance with 11.2 of IEC 61169-1:2013 1) Information prescribed in 11.1 of IEC 61169-1:2013 detailed above 2) Nominal characteristic impedance Ω 3) Assessment level code letter 4) Any additional marking required
– Ordering information: 1) Number of the detail specification /variant code 2) Assessment level code letter 3) Body finish (if more than one listed) 4) Any additional information or special requirements
– Related documents (if not included in IEC 61169-1:2013 or sectional specification): .
– Structural similarity in accordance with 10.2.2 of IEC 61169-1:2013
Relevant information on a basic style should be entered as variant 01.

6 Marking

6.1 Marking of component

Each component shall be legibly and durably marked, where space permits and in the following order of precedence, with:

- a) identity code of the manufacturer;
- b) manufacturer's connector identification code or IEC connector designation.

6.2 Marking and contents of package

The package shall be marked with the information prescribed in 6.1 and, in addition, the following information shall be given:

- a) nominal characteristic impedance;
- b) manufacturing date code;

c) any additional marking required by the relevant specification.

When required by the relevant specification, the package shall also include instructions for assembling the connector(s) and instructions for the use of any special tools or materials, as necessary.

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